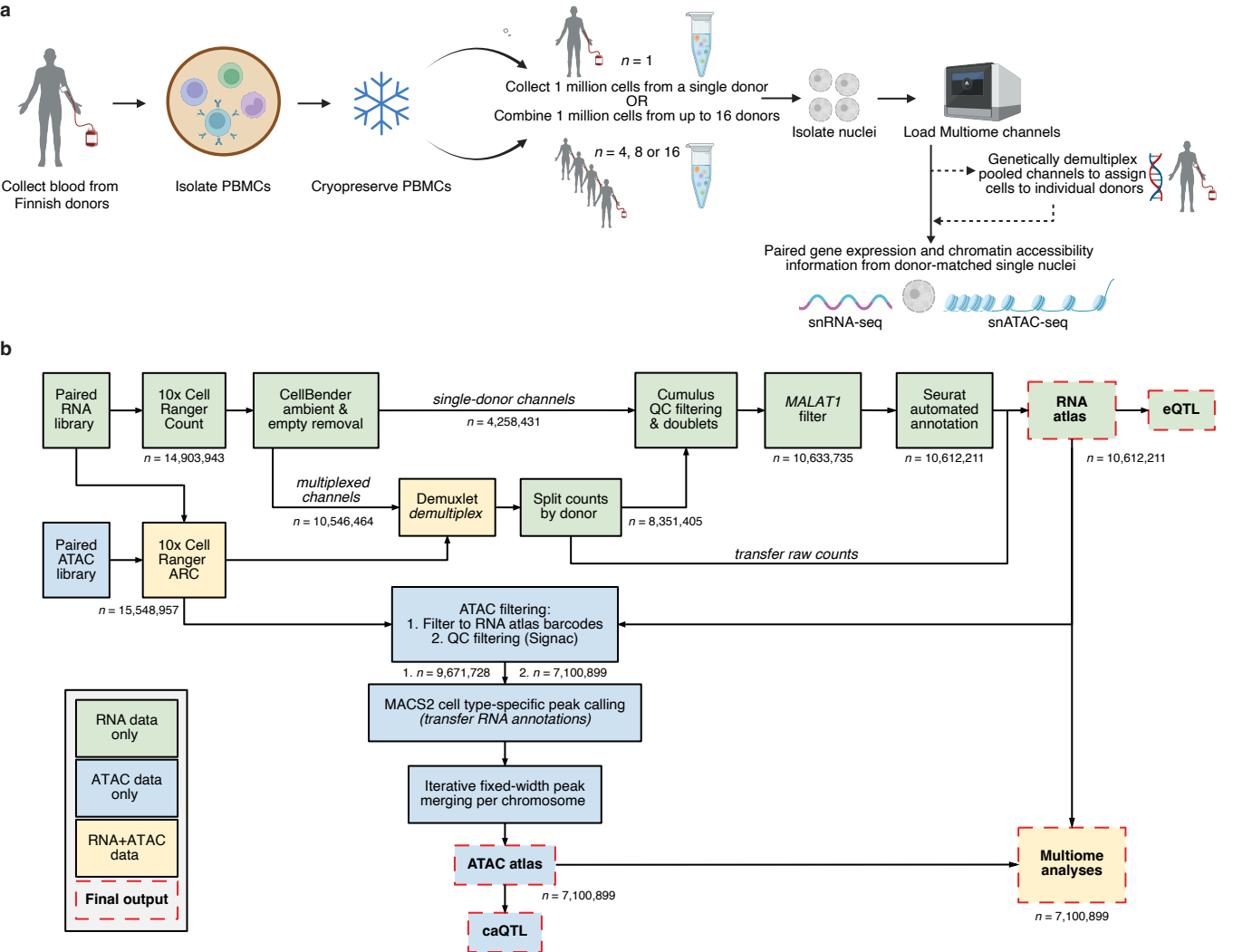
a
Collect blood from Finnish donors
Isolate PBMCs
Cryopreserve PBMCs
n = 1
Collect 1 million cells from a single donor
OR
Combine 1 million cells from up to 16 donors
n = 4, 8 or 16
Isolate nuclei
Load Multiome channels
Genetically demultiplex pooled channels to assign cells to individual donors
Paired gene expression and chromatin accessibility information from donor-matched single nuclei
snRNA-seq
snATAC-seq
b
Paired RNA library
10x Cell Ranger Count
n = 14,903,943
CellBender ambient & empty removal
single-donor channels
n = 4,258,431
Cumulus QC filtering & doublets
MALAT1 filter
n = 10,633,735
Seurat automated annotation
n = 10,612,211
RNA atlas
n = 10,612,211
eQTL
multiplexed channels
n = 10,546,464
Demuxlet demultiplex
Split counts by donor
n = 8,351,405
transfer raw counts
Paired ATAC library
10x Cell Ranger ARC
n = 15,548,957
ATAC filtering:
1. Filter to RNA atlas barcodes
2. QC filtering (Signac)
1. n = 9,671,728
2. n = 7,100,899
MACS2 cell type-specific peak calling
(transfer RNA annotations)
Iterative fixed-width peak merging per chromosome
ATAC atlas
n = 7,100,899
caQTL
Multiome analyses
n = 7,100,899
RNA data only
ATAC data only
RNA+ATAC data
Final output